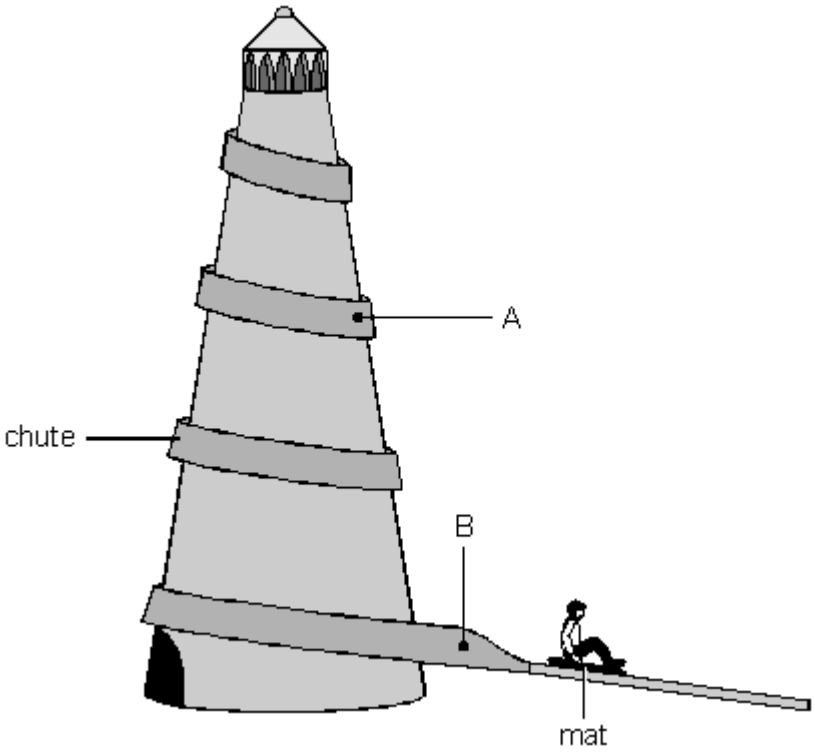
Q1. Anil sits on a mat at the top of a helter-skelter and then slides down a chute around the outside.



(a) (i) Name **two** of the forces acting on Anil as he slides from point A to point B.

1. .....

2. .....

2 marks

(ii) As Anil slides from point A to point B, the forces acting on him are balanced.

Describe Anil's speed when the forces acting on him are balanced.

.....

1 mark

(b) Anil goes back for a second go. This time he sits on a smooth cushion instead of a mat.

He goes much faster on the cushion. Give the reason for this.

.....

1 mark

(c) On his third go Anil lies back on the cushion with his arms by his side.

What happens to his speed? Give the reason for your answer.

.....

.....

.....

2 marks
Maximum 6 marks

Q1.

(a) (i) any **two** from

- gravity **or** weight
- friction
- reaction
  accept 'upthrust'
- air resistance
  accept 'drag'
  *do **not** accept 'centrifugal force'*
  ***or** 'centripetal force' **or** 'g- force'*

2 (L6)

(ii) any **one** from

- constant speed
- steady speed
- it stays the same
  *accept 'it is the same' **or** 'it does not change'*

1 (L6)

(b) friction is less
*'it is smoother' **or** 'it is slippery' are insufficient*

1 (L5)

(c) it increases
accept 'he goes more quickly'

1 (L6)

because there is less air resistance **or** friction
*accept 'he is streamlined **or** aerodynamic'*

1 (L6)

[6]